2036120007

ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

ENGINEERING MATHEMATICS

B.E - FOURTH SEMESTER (CS, DS, AM and IT)

I- Mid Term Test

CSBS401/DSBS401/ITBS401/AMBS401- DISCRETE MATHEMATICS

Time: 90 Minutes | Max: 40 Marks

PART- A (8x 1 =8 Marks)
Answer ALL Questions

		Marks	CO	BTL
1.	*What is mean by tautology?*	1	co1	K1
2.	*Define Rule P and Rule CP in the inference theory*	1	co1	K1
3.	*Define Tautological Implication*	1	co1	K1
4.	*State Absorption laws in logic*	1	co1	K1
5.	*If A={1} and B={1,2} find B×A.*	1	co2	K1
6.	*Define subset of a set. What is mean by proper subset?*	1	co2	K1
7.	*If A={a} and B={a,b} find A-B*	1	co2	k1
8.	*Define disjoint sets with a example*	1	co2	k1

PART- B (4 x 3 = 12 Marks)
Answer either (a) or (b) in Each Question

9.	**(a)**	*Prove that* $P \to (Q \to R) \Leftrightarrow (P \wedge Q) \to R$	3	co1	K2
		(OR)			
	(b)	*Prove that* $\neg Q \wedge (P \to Q) \Rightarrow \neg P$	3	co1	K2
10	**(a)**	*Find whether the conclusion C:* $\neg P$ *follows from the premises* $H_1: \neg(P \wedge \neg Q)$; $H_2: \neg R$; $H_3: Q \to R$ *using truth table technique*	3	co1	K3
		(OR)			
	(b)	*Prove that* $(P \to Q) \wedge (Q \to R) \Rightarrow (P \to R)$	3	co1	K3
11	**(a)**	*If A,B and C are any three sets Prove that* $A - (B \cup C) = (A - B) \cap (A - C)$	3	co2	K2
		(OR)			
	(b)	*If A,B and C are any three sets Prove that* $(A \cap B) - C = (A - C) \cap (B - C)$	3	co2	K2

33/2318 31/2317 29/2316 33/2315 37/2213 16/2212 31/2212

12 **(a)** *If R is the relation on* $A = \{1,2,3\}$ *such that* $(a,b) \in R$, *if and only if* $a + b = even\ number$, *find the relational matrix* M_R *hence find* R^{-1} *and* R^2 — 3 CO2 K3

(OR)

(b) *If R and S be relations on a set A={1,2,3} represented by the matrices* $M_R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ *and* $M_S = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ *find* $R \circ S,\ S \circ R$ *and* $R \oplus S$ — 3 CO2 K3

PART- C (2 x 10 = 20 Marks)
Answer either (a) or (b) in Each Question

13 **(a)** *Show that* $R \to S$ *can be derived from the premises* $P \to (Q \to S)$, $\neg R \vee P$ *and* Q — 10 Co1 K4

(OR)

(b) *Obtain the PDNF and PCNF of* $(\neg P \to \neg R) \wedge (P \leftrightarrow Q)$ — 10 Co1 K4

14 **(a)** *If R is the relation on the set of ordered pairs of positive integers such that* $(a,b)\ R\ (c,d)$ *if and only if* $ad = bc$ *Show that R is an equivalence relation* — 10 Co2 K4

(OR)

(b) *Prove that* $(A \oplus B) \times C = (A \times C) \oplus (B \times C)$ — 10 Co2 K4